

Geometric Heuristics for Basis Selection in the GSVD Method of Particular Solutions

Abstract

We give a closed-form heuristic system for choosing the type, count, and placement of particular solutions in the GSVD formulation of the Method of Particular Solutions (MPS) for the planar Dirichlet Laplace eigenproblem. The basis consists of corner-adapted fractional-order Fourier–Bessel functions $J_{k\alpha}(\sqrt{\lambda}r)\sin(k\alpha\theta)$ together with real-valued mono- and multipole fundamental solutions $Y_k(\sqrt{\lambda}r)\{\sin, \cos\}(k\theta)$ placed at exterior source points. All prescriptions depend only on measurable geometric data — area, perimeter, diameter, corner angles, and inter-corner distances — and on a single accuracy parameter $\Lambda = \ln(1/\hat{\epsilon})$ derived from the target eigenvalue precision via the Moler–Payne bound.

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The system is built around two error mechanisms that behave completely differently — *oscillatory resolution*, which scales like $\kappa = \sqrt{\lambda}$, and *analytic-continuation decay*, which scales like $\log(1/\epsilon)$ — plus a *partition-of-responsibility* principle that decides which basis family owns which piece of $\partial\Omega$. Everything below reduces to closed-form counts in terms of $A = |\Omega|$, $L = |\partial\Omega|$, $D = \text{diam } \Omega$, the corner angles ω_c , and inter-corner distances.

1 Notation, and converting ϵ into a target tension

Let $\kappa = \sqrt{\lambda}$ and $\kappa_{\max} = \sqrt{\lambda_{\max}}$. Corner c has vertex v_c , interior angle ω_c , and corner exponent

$$\alpha_c = \pi/\omega_c, \quad \text{basis } J_{j\alpha_c}(\kappa r_c) \sin(j\alpha_c\theta_c), \quad j = 1, \dots, M_c.$$

The tension $t = \|u\|_{L^2(\partial\Omega)} / \|u\|_{L^2(\Omega)}$ has units of $\text{length}^{-1/2}$, so form the dimensionless quantity

$$\hat{t} = \sqrt{A/L} \, t, \quad \frac{|\lambda - \lambda_n|}{\lambda_n} \lesssim C_\Omega \hat{t}.$$

Here C_Ω is $O(1)$ – $O(10)$ for non-pathological domains. **Calibrate it once** per domain family against a known eigenvalue rather than trusting a literature constant. Then set the design tolerance

$$\boxed{\hat{\epsilon} = \frac{\epsilon}{10 C_\Omega}, \quad \Lambda := \ln(1/\hat{\epsilon}).} \tag{1}$$

The parameter Λ is the single “how many digits” quantity that appears in every count below. Design for the *linear* Moler–Payne relation, not for the quadratic behaviour that is sometimes observed empirically — if the quadratic bonus materialises, it is free margin.

Hard floor. In IEEE double precision, $\sigma_{\min}$ of the GSVD pencil plateaus around 10^{-13} – 10^{-14} even with column equilibration, so $\epsilon \lesssim 10^{-12}$ is the practical limit. Below that the remedy is extended precision, not more basis functions.

2 The organizing principle

Two distinct things limit a particular-solution basis.

(a) Oscillation.

Any trace on $\partial\Omega$ has arclength wavenumber up to κ . The function $J_\nu(\kappa r)$ is exponentially small once $\nu > \kappa r$, with an Airy transition of width $\sim (\kappa r)^{1/3}$. So a corner expansion reaching radius R needs orders up to $\nu \approx \kappa R + 2(\kappa R)^{1/3}$.

(b) Analytic continuation.

A Fourier–Bessel series about v_c converges like $(r/d_c)^\nu$, where d_c is the distance from v_c to the nearest *singularity of the analytic continuation of u* — not the nearest corner. Beyond $r = d_c$ the series is useless no matter how many terms are added.

This is why single-corner MPS works spectacularly on the L-shape — all the other corners are $\pi/2$, hence regular, hence not singularities of the continuation — and fails on domains with several reentrant corners.

Design Principle 1 (Partition of responsibility). *Each singular corner owns only the boundary arc within radius*

$$R_c = \gamma d_c, \quad \gamma \in [0.3, 0.5],$$

and the MFS sources own everything else. Do not ask a corner expansion to cover the whole boundary unless $d_c = \infty$.

The *obstruction set* used to compute d_c is

$$\begin{aligned} \mathcal{S}^* = & \{\text{singular corners}\} \cup \{\text{their reflections across each edge}\} \\ & \cup \{\text{Schwarz-function singularities of curved arcs}\}, \end{aligned}$$

with the practical proxy $2 \times$ (distance to the nearest non-adjacent edge) for the reflected images, and the radius of curvature for smooth curved arcs. Then

$$d_c = \text{dist}(v_c, \mathcal{S}^* \setminus \{v_c\}),$$

capped by the largest radius actually needed, $d_c \leftarrow \min(d_c, \max_{x \in \partial\Omega} |x - v_c|/\gamma)$.

There is a genuine optimum in γ : raising it shrinks the MFS workload but inflates M_c through both mechanisms. The total is quite flat over $\gamma \in [0.3, 0.5]$; use $\gamma = 0.4$ unless you profile.

3 Corner classification: which corners get enrichment?

Compute $\mu_c = \text{dist}(\alpha_c, \mathbb{Z})$.

Case	Condition	Treatment
Regular	$\mu_c = 0$ (i.e. $\omega_c = \pi/m$)	No singular enrichment; u extends by reflection.
Weakly singular	$\mu_c > 0, \alpha_c > \Lambda/(2 \ln 15)$	Treat as regular.
Singular	$\mu_c > 0, \alpha_c \leq \Lambda/(2 \ln 15)$	Full corner-adapted treatment (§4).
Near-integer	$0 < \mu_c \lesssim 0.05$	Singular, but scale the continuation term by $\min(1, \mu_c/0.05)$.

The weakly singular cutoff comes from the observation that a smooth basis resolves r^{α_c} to $\hat{\epsilon}$ with degree $\sim \hat{\epsilon}^{-1/(2\alpha_c)}$, which is single digits when α_c is large. For $\hat{\epsilon} = 10^{-11}$ ($\Lambda \approx 25$) the cutoff is $\alpha_c \lesssim 4.6$: **corners wider than about 39° get corner-adapted terms; sharper spikes do not.**

Should regular corners receive corner-adapted terms? Not a Λ -sized budget. Two nuances:

- A *small* set of $J_{jm}(\kappa r) \sin(jm\theta)$ at a regular corner ($m = \pi/\omega_c$ integral) is cheap and often improves conditioning, because it supplies well-localised angular content that the MFS represents only awkwardly. Use $M_c = \lceil \frac{\omega_c}{\pi} (\kappa R_c^{\text{loc}} + 5) \rceil$ with R_c^{loc} equal to half the shorter adjacent edge, and no continuation term. Treat this as optional.
- The operationally important difference is in the MFS: **do not taper the source offset to zero at a regular corner** (§5). That is where most of the savings live.

4 Corner-adapted counts

For each singular corner,

$$M_c = \underbrace{\left\lceil \frac{\omega_c}{\pi} (\kappa_{\max} R_c + 2(\kappa_{\max} R_c)^{1/3} + 5) \right\rceil}_{\text{oscillatory}} + \underbrace{\left\lceil \frac{\omega_c}{\pi} \cdot \frac{\Lambda}{\ln(1/\gamma)} \right\rceil}_{\text{continuation}} \quad (2)$$

Note the structure: the angle enters *only* through the factor $\omega_c/\pi = 1/\alpha_c$, because the index spacing in ν is α_c . Wider corners cost proportionally more terms for the same $\nu_{\max}$ — a $3\pi/2$ reentrant corner costs three times what a $\pi/2$ corner would at equal reach. That single factor is the whole angle dependence; there is no separate “reentrant corners are harder” term, because the difficulty is already captured by $1/\alpha_c$ and by d_c shrinking when singular corners cluster.

Conditioning cap. Truncate at

$$\nu_{\max}^{(c)} \leq \kappa R_c + \frac{\ln(1/\epsilon_{\text{mach}})}{\ln(R_c/r_{\text{mid}})}, \quad r_{\text{mid}} = \text{median } |x - v_c| \text{ over the owned arc,}$$

since beyond that the column’s dynamic range over the sampled boundary exceeds what double precision can carry, and the GSVD will discard it anyway. Always **column-equilibrate** (unit norm with respect to the combined boundary + interior quadrature) before forming the pencil; without it the J_ν and Y_k columns differ by hundreds of orders of magnitude.

5 Fundamental-solution placement

Treat the offset sources and the lightning clusters as *one graded curve*, not two families. That is the cleanest way to keep them consistent.

5.1 Local offset

For $s \in \partial\Omega$,

$$\delta(s) = \min \left(\delta_{\text{amb}}, \eta \cdot \text{dist}(s, \mathcal{S}^*) \right), \quad \eta \approx 0.3, \quad \delta_{\text{amb}} = \min \left(0.25D, \frac{1}{3}\rho_{\text{curv}}^{\min} \right).$$

Sources go at $s + \delta(s)n(s)$ (outward normal). Because $\mathcal{S}^*$ contains only *singular* corners and their images, δ stays at δ_{amb} across regular corners and collapses only at singular ones — exactly the geometric clustering one wants, derived rather than imposed.

5.2 Local spacing

The classical MFS/Fourier result for a source curve at distance δ with spacing h is boundary error $\sim e^{-\pi\delta/h}$. Hence

$$h(s) = \min \left(\frac{2\pi}{3\kappa_{\max}}, \frac{\pi\delta(s)}{\Lambda} \right) \quad (3)$$

The first argument is the Nyquist / three-points-per-wavelength floor; the second is the accuracy requirement. For high-accuracy work the second almost always binds, meaning **MFS sources are oversampled several-fold relative to Nyquist** — that is normal, and is the price of the offset.

5.3 Total count

$$N_{\text{mfs}} \approx \max\left(\frac{3\kappa_{\text{max}}L}{2\pi}, \frac{\Lambda L}{\pi\delta_{\text{amb}}}\right) + \sum_{c \text{ sing}} n_c^{\text{clus}}.$$

Since $h(s) \propto \text{dist}(s, v_c)$ near a singular corner, the tapered region contributes geometrically clustered sources with ratio $1 + \pi\eta/\Lambda$ per step; the count for the tail is $n_c^{\text{clus}} \approx \ln(\delta_{\text{amb}}/s_{\text{min}})/\ln(1 + \pi\eta/\Lambda)$.

5.4 Where to stop the taper

This is where the Fourier–Bessel basis and the clusters must not duplicate each other.

- **Corner has FB terms** (§4): stop the taper at $s_{\text{min}} \approx 0.05 R_c$ and keep only a short tail, $n_c^{\text{clus}} \approx 8\text{--}12$ poles on the outward angle bisector. Their job is to bridge the region where the FB series is still converging but the ambient MFS cannot reach — a conditioning bridge, not an approximation tool. More than this is wasted and actively hurts σ_{min} .
- **Corner has no FB terms** (weakly singular by §3, or M_c capped for conditioning): use full Gopal–Trefethen tapered clustering on the outward bisector,

$$s_j = \delta_{\text{amb}} e^{-\sigma(\sqrt{n_c} - \sqrt{j})}, \quad \sigma \approx 4, \quad j = 1, \dots, n_c,$$

with root-exponential convergence $\sim e^{-\pi\sqrt{2\alpha_c n_c}}$, giving

$$\boxed{n_c^{\text{clus}} \approx \frac{\Lambda^2}{2\pi^2\alpha_c}} \tag{4}$$

5.5 Multipole orders

Use plain monopoles (Y_0) for the clustered near-corner sources — they are close to the boundary and low order suffices. Reserve multipole stacks for a handful of far centres, where they permit far fewer sources. For a centre at distance ρ_j from the arc it serves, of half-extent a_j ,

$$K_j \approx \kappa\sqrt{\rho_j^2 + a_j^2} + \frac{\Lambda}{\ln(\sqrt{\rho_j^2 + a_j^2}/\rho_j)}, \quad K_j \leq \frac{\ln(1/\epsilon_{\text{mach}})}{\ln(1 + D/\rho_j)}.$$

The second inequality is a hard conditioning cap: $Y_k(\kappa r) \sim -\frac{(k-1)!}{\pi}(2/\kappa r)^k$ has dynamic range $(r_{\text{max}}/r_{\text{min}})^K$ across Ω , and once that exceeds $1/\epsilon_{\text{mach}}$ the column carries no information.

Use only Y_k at exterior centres, never J_k — the regular part about an exterior centre is already spanned by the interior and corner Fourier–Bessel functions and merely duplicates them.

6 Collocation, the A_I block, and quadrature

Replace the interior points. With interior sampling, M_I must scale like $A\kappa^2$ to control $\|u\|_{L^2(\Omega)}$ reliably, and undersampling produces spurious tension minima that look exactly like eigenvalues. Use the Rellich identity instead: for a Dirichlet solution of $\Delta u + \lambda u = 0$,

$$\|u\|_{L^2(\Omega)}^2 = \frac{1}{2\lambda} \int_{\partial\Omega} (x \cdot n) |\partial_n u|^2 ds, \tag{5}$$

with x measured from any origin — choose one making $x \cdot n > 0$, i.e. a star centre, if Ω is star-shaped. Then *both* blocks of the pencil are boundary matrices, M_I becomes $O(\kappa)$ rather than $O(\kappa^2)$, and the norm-equivalence constants relating σ_{min} to the true tension become κ -independent — which matters, since the whole $\epsilon \mapsto \hat{\epsilon}$ chain of §1 assumes it.

Boundary nodes. Take $M_B \approx 3N$ in total, with weights folded in as $\sqrt{w_i}$ so the discrete norms actually approximate L^2 ; otherwise the topological-equivalence constant drifts with κ and the calibration of C_Ω is meaningless. Use composite Gauss–Legendre panels, geometrically refined toward each singular corner with ratio $1/2$, down to $\approx s_{\min}$ from §5 — roughly $\log_2(\ell_e/s_{\min})$ levels per adjacent edge.

7 Total budget and precision floors

$$N_{\text{tot}} = \sum_{c \text{ sing}} M_c + \sum_{c \text{ reg}} M_c^{\text{opt}} + N_{\text{mfs}}.$$

Sanity checks before building the matrix:

- If $N_{\text{tot}} \gtrsim 600$ in double precision, conditioning will limit the GSVD before approximation power does. Relax ϵ , reduce γ , or move to extended precision; adding functions will make $\sigma_{\min}$ *worse*.
- N_{tot} should scale like $O(\kappa_{\max}L) + O(\Lambda \cdot n_{\text{sing}})$, i.e. like $\sqrt{\lambda_{\max}}$, not $\lambda_{\max}$. If the heuristic produces $O(\lambda)$ growth, something is wrong — usually interior points.
- If $\sigma_{\min}$ stops decreasing as N increases, you are on the floor; enrichment will not help.

8 λ -sweep bookkeeping

Every count above depends on κ . **Do not** recompute the basis continuously during a sweep: discontinuous basis changes put kinks in $\sigma_{\min}(\lambda)$ and wreck root-finding and minimum tracking. Freeze the basis on dyadic windows $\lambda \in [\lambda_0, 2\lambda_0)$, sized at the window’s upper end, with hysteresis at the window boundaries. Within a window $\sigma_{\min}(\lambda)$ is then smooth and the usual scaling and minimisation machinery applies.

9 Adaptive correction loop

The heuristic gets within a factor of roughly 1.5; close the loop.

1. Build with the counts above; compute $\sigma_{\min}$ at a converged eigenvalue.
2. If $\hat{\epsilon}$ is not met, evaluate the boundary residual $|u(s)|$ panel by panel.
3. Attribute the largest residual to its owner: inside R_c of a singular corner, increase M_c by 25%; outside, halve h on that arc, or — if δ there is already at $\eta \text{dist}(s, \mathcal{S}^*)$ — add a taper level.
4. If the residual is *uniform* along $\partial\Omega$, you are conditioning-limited, not resolution-limited: increase δ_{amb} *and* decrease h together (keeping $\pi\delta/h = \Lambda$), or drop the weakest GSVD directions.

10 Worked example: the L-shape

Take $\lambda \approx 100$, $\epsilon = 10^{-10}$; $A = 3$, $L = 8$, $D = \sqrt{5}$, $\kappa = 10$. With $C_\Omega = 10$ we get $\hat{\epsilon} = 10^{-11}$ and $\Lambda = 25.3$. The corners are one at $3\pi/2$ ($\alpha = 2/3$, singular) and five at $\pi/2$ (regular).

- Only one singular corner, so $\mathcal{S}^*$ reduces to its own reflections: the nearest non-adjacent edge is at distance 1, giving $d_c = 2$. With $\gamma = 0.4$, $R_c = 0.8$.

- Oscillatory term: $\kappa R_c = 8$, so $\nu = 8 + 4.0 + 5 = 17$ and $M^{\text{osc}} = 17 \times 1.5 = 26$.
- Continuation term: $\nu = 25.3 / \ln 2.5 = 27.6$, so $M^{\text{cont}} = 41$. Hence $M_c = 67$.
- MFS: $\delta_{\text{amb}} = 0.25D = 0.56$, so $h = \pi(0.56)/25.3 = 0.070$ — the Nyquist floor would be 0.21, so accuracy binds. Then $N_{\text{mfs}} \approx 8/0.070 \approx 115$, plus a ~ 10 -pole bridge at the reentrant corner.
- Regular corners: no taper and no continuation budget — δ stays at 0.56 all the way around them.

Totals: $N_{\text{tot}} \approx 195$, $M_B \approx 600$ graded boundary nodes, no interior points. That is the right order for ten digits at $\lambda \approx 100$ on this domain.

11 Things to verify in a given implementation

Source-curve resonance.

Real MFS bases built from Y_k on a closed exterior curve Γ' can lose completeness at isolated λ related to the spectrum of the region bounded by Γ' . It manifests as an isolated spike in $\sigma_{\min}$ that moves when Γ' is perturbed. Cheap diagnostic: re-run suspicious λ with δ_{amb} scaled by 1.3; genuine eigenvalues do not move. Including multipole orders mitigates this substantially.

C_Ω calibration

is the weakest link in the chain — but everything downstream is $\log(1/\epsilon)$, so an order-of-magnitude error in C_Ω costs only about 10% more basis functions. Do not over-engineer it.

Degeneracies.

With multiplicity greater than one, a single $\sigma_{\min}$ is not enough: track the m smallest generalized singular values and use the tension of the whole subspace. The counts above are unchanged.

Tuned versus derived constants.

The values $\gamma = 0.4$, $\eta = 0.3$, $\sigma = 4$, $M_B/N = 3$, and the $2(\kappa R)^{1/3} + 5$ transition margin are tuned, not derived. The structures $\Lambda/\ln(1/\gamma)$, $\pi\delta/h$, $\Lambda^2/(2\pi^2\alpha)$, and ω_c/π are derived and should be trusted more than the constants in front of them.